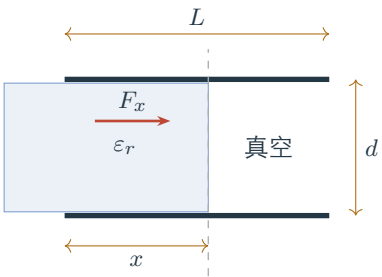
M3-028 固定电荷下介质插入的能量与功



纸面外宽 w；$+x$ 沿插入方向向右

$$C_0 = \varepsilon_0 w L/d, \quad C_L = \varepsilon_r C_0$$

$$\Delta U = -\frac{Q^2}{2C_0}(1 - 1/\varepsilon_r)$$

$$W_{\text{field}} = -\Delta U > 0$$